

# IMO 2026 — Problem 2

Solution by GPT-5.6 Sol (lightweight harness)

2026-07-18

**Source:** `results/imo-2026/gpt-5.6-sol-lightweight/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

## Problem

Let  $ABC$  be a triangle and let points  $M$  and  $N$  be the midpoints of sides  $AB$  and  $AC$ , respectively. Let points  $K$  and  $L$  be chosen inside triangles  $BMC$  and  $BNC$ , respectively, such that  $K$  lies inside the angle  $LBA$ ,  $L$  lies inside the angle  $ACK$ , and  $\angle KBA = \angle ACL$ ,  $\angle LBK = \angle LNC$ ,  $\angle LCK = \angle BMK$ . Let  $O$  be the circumcentre of triangle  $AKL$ . Prove that  $OM = ON$ .

## Solution document

### Status

solved

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### Approaches tried

- Trigonometric/complex coordinates: successful. Coordinates of  $K, L$  follow from the sine rule in  $BMK, CNL$ ; the two remaining angle conditions become reality relations.
- Equal powers: successful. A standalone complex-algebra lemma (`lemmas/complex-algebra.md`) shows that those reality relations imply that  $M, N$  have equal powers with respect to  $(AKL)$ .
- Symbolic Laurent-polynomial checks (`code/symbolic.py`, `code/eliminate.py`, `code/certificate.py`) were used to check the algebra and orientations. An apparent orientation correction was rechecked: for counterclockwise  $ABC$ , the ray directions are  $CL = \alpha + \pi + x$  and  $CK = \alpha + \pi + x + z$ .

### Current best

The claim follows by normalizing  $A = 0, B = 2, C = 2de^{i\alpha}$ . Then  $M = 1, N = de^{i\alpha}$ , while the sine rule gives explicit complex coordinates for  $K, L$ . The angle order assumptions turn the remaining conditions into exactly the two hypotheses of the complex equal-power lemma, proving that  $M, N$  have equal powers with respect to  $(AKL)$ . As both powers equal the squared distance to  $O$  minus the same circumradius squared,  $OM = ON$ .

### Full proof

Reflect the whole configuration if necessary, and assume that  $A, B, C$  occur counterclockwise. We use complex coordinates, normalized by

$$A = 0, \quad B = 2, \quad C = 2n, \quad n = de^{i\alpha},$$

where  $d > 0$  and  $\alpha = \angle BAC$ . Thus

$$M = 1, \quad N = n.$$

Set

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK,$$

and abbreviate

$$X = e^{ix}, \quad Y = e^{iy}, \quad Z = e^{iz}, \quad a = e^{i\alpha}.$$

We first find  $K$ . In triangle  $BMK$  we have

$$BM = 1, \quad \angle MBK = x, \quad \angle BMK = z.$$

Consequently  $\angle BKM = \pi - x - z$ , so the sine rule gives

$$MK = \frac{\sin x}{\sin(x+z)}.$$

The ray  $MK$  makes angle  $z$  with the positive real axis; hence

$$K = 1 + \frac{\sin x}{\sin(x+z)}Z. \quad (1)$$

Likewise, in triangle  $CNL$  we have  $CN = d$ ,  $\angle NCL = x$ , and  $\angle CNL = y$ . The ray  $NL$  has direction  $\alpha - y$ , and the sine rule gives

$$L = n \left( 1 + \frac{\sin x}{\sin(x+y)}Y^{-1} \right). \quad (2)$$

The denominators in (1)–(2) are nonzero because they are sines of angles supplementary to angles of the nondegenerate triangles  $BMK, CNL$ .

We next translate the ordering of the rays. Since  $K$  lies inside  $\angle LBA$ , when one turns inside that angle from  $BA$  toward  $BL$ , one meets  $BK$  first. Therefore  $BL$  has direction  $\pi - x - y$ . It follows that

$$(L-2)XY \in \mathbb{R}. \quad (3)$$

At  $C$ , the ray  $CA$  has direction  $\alpha + \pi$ . Since  $\angle ACL = x$ , the ray  $CL$  has direction  $\alpha + \pi + x$ . Since  $L$  lies inside  $\angle ACK$  and  $\angle LCK = z$ , the ray  $CK$  has direction  $\alpha + \pi + x + z$ . Thus the opposite vector  $C - K = 2n - K$  has direction  $\alpha + x + z$ , and hence

$$\frac{2n - K}{aXZ} \in \mathbb{R}. \quad (4)$$

We now apply the complex equal-power lemma proved in `lemmas/complex-algebra.md`; its short algebraic proof is reproduced here for self-containment. If  $p, q$  denote the complex coordinates of  $K, L$ , respectively, then, since

$$\frac{\sin x}{\sin(x+z)} = \frac{X - X^{-1}}{XZ - X^{-1}Z^{-1}}, \quad \frac{\sin x}{\sin(x+y)} = \frac{X - X^{-1}}{XY - X^{-1}Y^{-1}},$$

formulas (1)–(2), together with (3)–(4), satisfy the hypotheses of that lemma. We recall its verification.

Write  $t^* = \bar{t}$ . The circle through  $0, p, q$  has equation

$$w\bar{w} + uw + v\bar{w} = 0. \quad (5)$$

Putting  $w = p, q$  and applying Cramer's rule gives, with  $\Delta = pq^* - qp^* \neq 0$ ,

$$u\Delta = -pp^*q^* + qq^*p^*, \quad v\Delta = -pqq^* + qpp^*. \quad (6)$$

The difference between the values of the left side of (5) at 1 and at  $n$  is, after multiplication by  $\Delta$ ,

$$H = (1 - nn^*)\Delta + (1 - n)(-pp^*q^* + qq^*p^*) + (1 - n^*)(-pqq^* + qpp^*). \quad (7)$$

The two reality relations (3)–(4) are equivalent to

$$(q - 2)XY = (q^* - 2)X^{-1}Y^{-1}, \quad (8)$$

$$(2n - p)a^{-1}X^{-1}Z^{-1} = (2n^* - p^*)aXZ. \quad (9)$$

Substitute

$$p = 1 + Z \frac{X - X^{-1}}{XZ - X^{-1}Z^{-1}}, \quad q = da \left( 1 + Y^{-1} \frac{X - X^{-1}}{XY - X^{-1}Y^{-1}} \right)$$

and their conjugates into (7). On using (8)–(9), and putting terms over the common denominator

$$(XZ - X^{-1}Z^{-1})^2(XY - X^{-1}Y^{-1})^2,$$

the numerator cancels in conjugate pairs: the four types are

$$(nq^*p^*, -n^*qp), \quad (npp^*q^*, -n^*pp^*q), \quad (qq^*p^*, -qq^*p), \quad (pq^*, -qp^*).$$

Thus  $H = 0$ . By (5)–(7), the powers of  $1 = M$  and  $n = N$  with respect to the circle through  $0, p, q$ , namely  $(AKL)$ , are equal.

Let  $R$  be the radius of  $(AKL)$ . The power of any point  $P$  with respect to this circle is  $OP^2 - R^2$ . Therefore

$$OM^2 - R^2 = ON^2 - R^2,$$

so  $OM^2 = ON^2$ . Both are nonnegative lengths, and hence

$$\boxed{OM = ON}.$$